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CLASS: AI & DS - 'A' - IIIrd year

1) Design an FSA for b^+

Step 1:
understand the Pattern
Pattern: b^+

Part	Meaning
b	The First b is compulsory
+	The Preceding b May repeat once (or) more times

Step 2: check accepted and rejected Strings

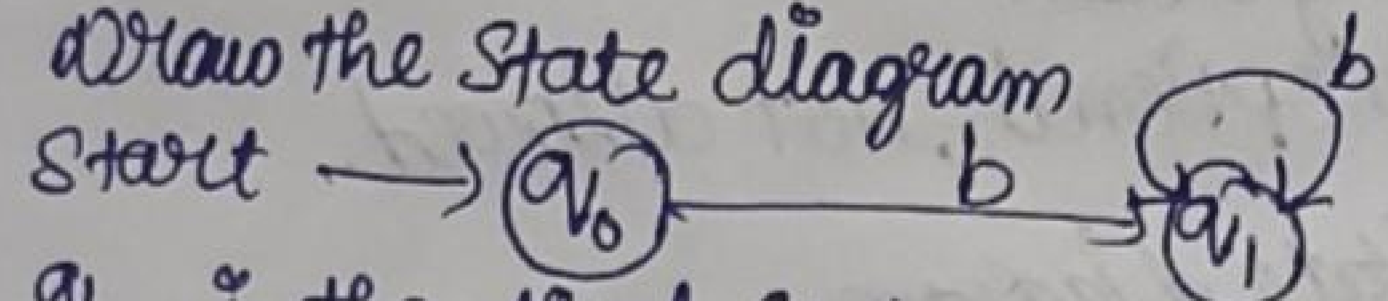
Accepted Strings	why accepted
b	contain 1 b
b b	contain 2 b's
b b b	contain 3 b's
b b b b	contain 4 b's

Rejected Strings	why rejected
ε (empty)	atleast one 'b' is required
a	The Symbol is not b
b a	contain a, not allowed
a b b	Start with a.

Step 3: Identify the Mistake

Mistake	Meaning
b^*	allow zero occurrence
ab^+	Start with a Not allowed
bb^+	extra b before first b

Step 4: Draw the State diagram



q_1 is the final State

Step 5: Transition table

current State	Input	Next State	Explanation
q_0	b	q_1	First b is read
q_1	b	q_1	extra b's are read

Step 6: Sample inputs
Accepted : b, bb, bbb, bbbb
Rejected : ε, a, ba, abb

Conclusion: The FSA accepts all strings having one (or) more occurrence of b.

2) Design an FSA for a?b

Step-1: understand the Pattern
Pattern: a?b

Part	Meaning
a	The a is optional
?	It may occur zero (or) one time
b	The b is compulsary.

Step 2: Accepted and Rejected Strings

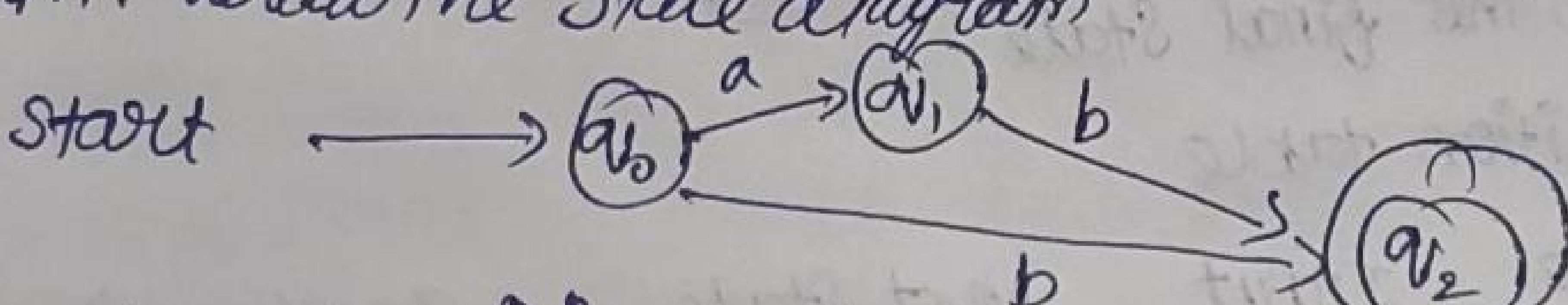
Accepted String	why accepted
b	a is absent
ab	a is present once

Rejected String	why Rejected
a	b is miss
bb	extra b
aa	b is missing
abb	extra b after ab

Step 3: Identify the Mistake.

a+?	More than one a
ba	wrong order
b?	b is optional, Not allowed.

Step 4: Draw the State diagram



Step 6: Transition table.

current State	Input	Next State	Explanation
q_0	a	q_1	reads optional a
q_0	b	q_2	b directly
q_1	b	q_2	after a, read b.

Step6: Sample input

Accepted : 01, 011, 0111, 01111

Rejected : 0, 10, 010

Conclusion: The FSA accepts strings starting with 0 followed by one (or) more 1's.

4) Design an FSA for cat/dog.

Step1: understand the pattern

Pattern: cat | dog

Part	Meaning
cat	accept the word cat
|	or operator
dog	accept the word dog

Step2: check accepted and rejected string

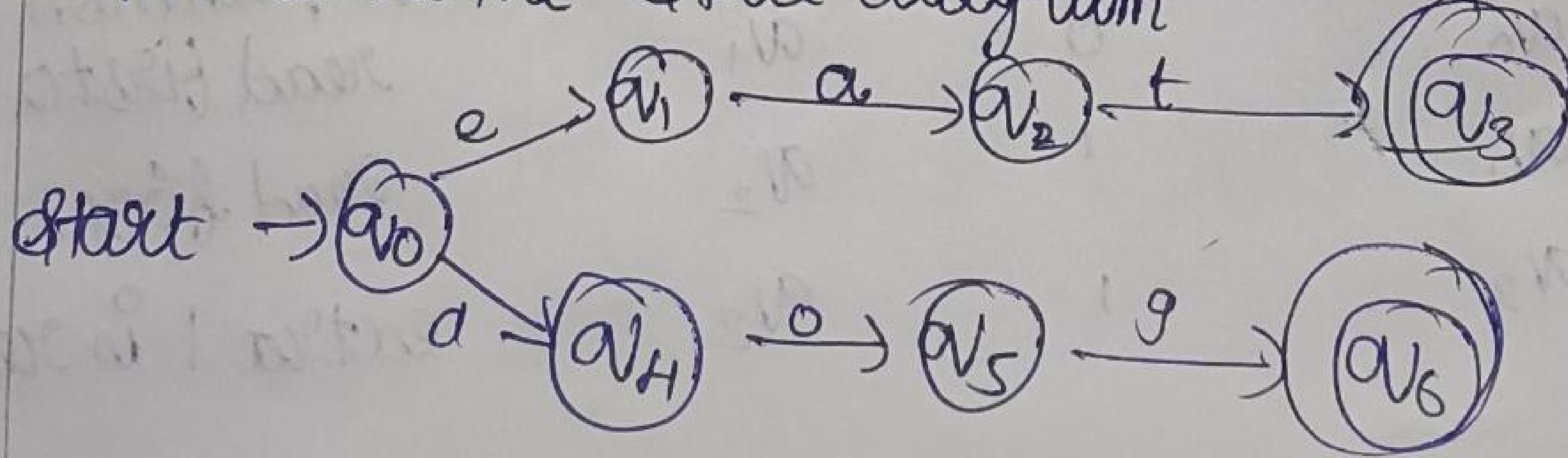
Accepted String	why accepted
cat	Matches cat
dog	Matches dog

Rejected strings	why rejected
ca	incomplete
do	incomplete
cats	extra character
dogs	extra character

Step3: Identify the mistakes.

cat	case sensitive [lower case]
dogg	extra g
catt	extra t

Step4: Draw the state diagram



Step6: Sample Inputs

Accepted: b, ab

Rejected : a, bb, aa, abb.

Conclusion: The FSA accepts either b (or) ab.

3) Design an FSA for 0,+

Step1: understand the Pattern

Pattern: 01+

Part	Meaning
0	The first Symbol Must be 0
1+	one (or) more 1's after 0

Step2: Check accepted and rejected strings

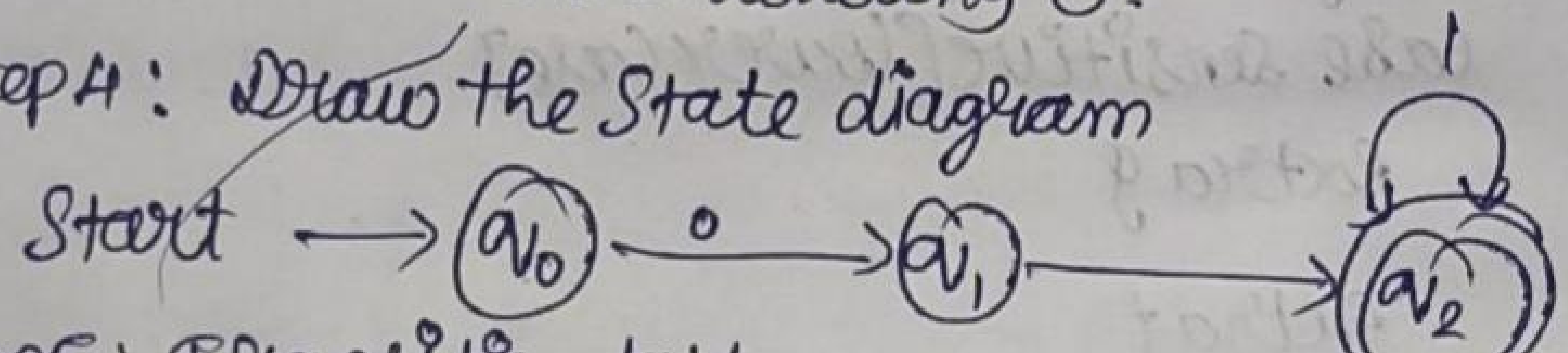
Accepted strings	why accepted
0 1	contain one 1
0 11	contain two 1's
0 111	contain three 1's

Rejected strings	why rejected
0	1 is missing
1	0 is missing
10	wrong order
010	ends with 0 Not allowed

Step3: Identify the Mistake

10	wrong order
01	allows zero ones Not allowed
001	extra leading 0.

Step4: Draw the State diagram



Step5: Transition table

current State	Input	Next State	Explanation
q_0	0	q_1	read first 0
q_1	1	q_2	read first 1
q_2	1	q_2	extra 1 is read.

Step 5:·

Transition table

Current State	Input	next State
q_0	c	q_1
q_1	a	q_2
q_2	t	q_3

Current State	Input	next State
q_0	d	q_4
q_4	o	q_5
q_5	g	q_6

Step→6

Sample inputs

Accepted : Cat, Dog

Rejected : Ca, do, catt, dogg

Conclusion:

The FSA accepts only the words cat (or) dog.